



Facilitating effects of Buyang Huanwu decoction on axonal regeneration after peripheral nerve transection

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ABSTRACT

Ethnopharmacological relevance: In traditional Asian medicine, Buyang Huanwu decoction (BYHWD) has been used for the treatment of cardiovascular and neurological disorders. Recent experimental studies have begun to provide evidence on the protective effects of BYHWD on injured peripheral nerves.

Aim of the study: To examine whether BYHWD was effective in inducing axonal regeneration after peripheral nerve transection, and if so, how it acted on the nerve.

Materials and methods: The sciatic nerve in rats was transected and resutured 0, 1, or 4 weeks later. BYHWD was orally administered daily into the animals with nerve transection and coaptation (NTC). Axonal regeneration was measured by immunofluorescence staining of NF-200 and superior cervical ganglion 10 (SCG10) and by retrograde tracing method. Changes of protein levels in the sciatic nerve were analyzed by western blot analysis. Effects of BYHWD and its constituents on neurite outgrowth were analyzed in cultured dorsal root ganglion (DRG) neurons. Hot plate and treadmill training tests were performed to assess the levels of functional recovery after nerve injury.

Results: The rate of axonal regeneration was attenuated by delayed coaptation after transection, but improved by BYHWD treatment. Levels of phospho-Erk1/2 and Cdc2 phosphorylation of vimentin, measured as indicators of the activation of regenerating axons and supportive Schwann cells, were increased in the sciatic nerve of NTC animals, and their distribution in the proximal and distal nerves were affected by BYHWD treatment. Treatment of BYHWD during the period of chronic denervation significantly increased axonal regeneration when analyzed by immunofluorescence staining and retrograde tracing methods. Neurite outgrowth of DRG neurons cocultured with Schwann cells from the chronically transected sciatic nerves was enhanced by BYHWD treatment. Radix Paeoniae Rubra induced neurite outgrowth most efficiently among all herbal constituents of BYHWD. Finally, hot plate and treadmill training tests demonstrated that BYHWD administration significantly improved the sensorimotor nerve function in NTC animals.

Conclusions: Our data suggest that BYHWD treatment may contribute to the timely interaction between regenerating axons and distal Schwann cells in the transected nerve and facilitate axonal regeneration.

1. Introduction

Although axons in the peripheral nerves have an intrinsic capacity to regrow after injury, the extent of regeneration and consequent functional recovery are affected by environmental factors, notably by Schwann cells in the injury area (Chen et al., 2007). After the nerve injury, Schwann cells undergo dedifferentiation followed by proliferation and redifferentiation through the interaction with regrowing axons and myelination (Namgung, 2014; Jessen et al., 2015). Unlike smaller sizes of experimental animals, the extent of axonal regeneration and

functional recovery is markedly limited in human patients suffering from peripheral nerve injury (Lee and Wolfe, 2000). One main reason for this is that the distance between the injury site and the targets is much longer in human than experimental animals, and thus, potentially supportive environmental factors including Schwann cells and extracellular matrix proteins in the distal degenerated nerve area may be deteriorated due to the lack of axonal interaction (Fu and Gordon, 1995; Höke, 2006).

The efficiency of axonal regeneration is more restrictive in the nerves after transection than crush injury. In order for a transected

Abbreviations: BYHWD, Buyang Huanwu decoction; DRG, dorsal root ganglion; NTC, Nerve transection and coaptation; PTSC, post transection and coaptation; PC, post coaptation; PTS, post transection; Cdc2, cell division cycle 2; DRG, dorsal root ganglion

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nerve to regenerate successfully, both ends of transected nerve need to be coapted and resutured properly since the regenerating axons should be extended through the basal lamina layer where Schwann cells are aligned underneath (Bunge et al., 1986; Cornbrooks et al., 1983). Here, a chronic denervation may worsen regenerative responses because of the possible inactivation of regenerating factors in the disconnected distal nerve due to the lack of interacting signals from regenerating axons (You et al., 1997; Boyd and Gordon, 2003). Overexpression of heat shock protein 27 (Hsp27) in mice accelerated axonal regeneration after prolonged nerve transection and led to the recovery of motor function, implicating potential application of Hsp27 to human patients (Ma et al., 2011). Thus, exploration of regeneration-related factors in the chronically denervated nerves may be of great importance to develop therapeutic strategies for nerve repair (Scherer and Salzer, 2001; Maggi et al., 2003).

In traditional Asian medicine, BYHWD has been used for the treatment of cardiovascular disorders including ischemic insults in neural tissues, and recent experimental studies have begun to provide supportive evidences (Pan et al., 2017; Yang et al., 2015; Wang et al., 2013; Li et al., 2003). It was also reported that BYHWD protected Schwann cells from oxidative stress and mediated regenerative responses after spinal cord injury (Chen et al., 2008; Yang et al., 2011). Recently, we have reported that BYHWD facilitated axonal regeneration in the sciatic nerve after crush injury by acting on regenerating axon as well as Schwann cells (Chang et al., 2016). While these studies indicate that BYHWD has a growth-promoting activity on injured axons, its effects on chronically denervated nerve have not been investigated. Having noted the potential importance of BYHWD on therapeutic application for peripheral nerve repair, we asked whether BYHWD was involved in facilitating axonal regeneration after nerve transection, and if so, how it worked. Our study demonstrates that BYHWD acts on both regenerating axons and Schwann cells, facilitates axonal regeneration, and improves functional recovery.

2. Materials and methods

2.1. Chemicals and herbal drug extraction

Dried herbal components of BYHWD were obtained from Daejeon University Korean Hospital. BYHWD was composed of the following herbal components: 120 g of Radix Astragali seu Hedysari (specimen number: KNP0011, production area: Jeongseon, Korea), 6 g of Radix Angelicae Sinensis (specimen number: KNP0008, production area: Jeongseon, Korea), 4.5 g of Radix Paeoniae Rubra (specimen number: KNP0085, production area: Younghae, Korea), 3 g of Pheretima (specimen number: KNP0053, production area: Jiangxi, China), 3 g of Rhizoma Ligustici Chuanxiong (specimen number: KNP0028, production area: Yeonghae, Korea), 3 g of Semen Persicae (specimen number: KNP0087, production area: Hebei, China), and 3 g of Flos Carthami (specimen number: KNP0020, production area: Xingang, Taiwan). The voucher specimens were deposited in the Herbarium of the Department of Korean Medicine, Daejeon University (Daejeon, Korea). Extraction procedure was essentially the same as reported previously (Chang et al., 2016). Briefly, a mixture of individual herbal components was chopped into small pieces and suspended in distilled water for 2 h. Then, they were boiled for 3 h and filtered with Whatman filter paper (Grade 1, Whatman Inc., Clifton, USA) three times, as previously reported (Zhang et al., 2010). The extract was frozen at -70°C for 4 h, and freeze-dried for 24 h. The yield for BYHW extract was 15–20% of the initial raw materials (dry weight). Final product was dissolved in physiological saline solution (1 g of extract residue/ml in 0.9% NaCl solution) and stored at -20°C . Chemical ingredients of BYHWD, amygdalin, calycosin, and kaempferol were all purchased from Sigma-Aldrich (St. Louis, USA), dissolved in saline and used to investigate their effects on neurite outgrowth. Similarly, individual herbal extracts of BYHWD were treated to cultured DRG neurons and used to determine neurite

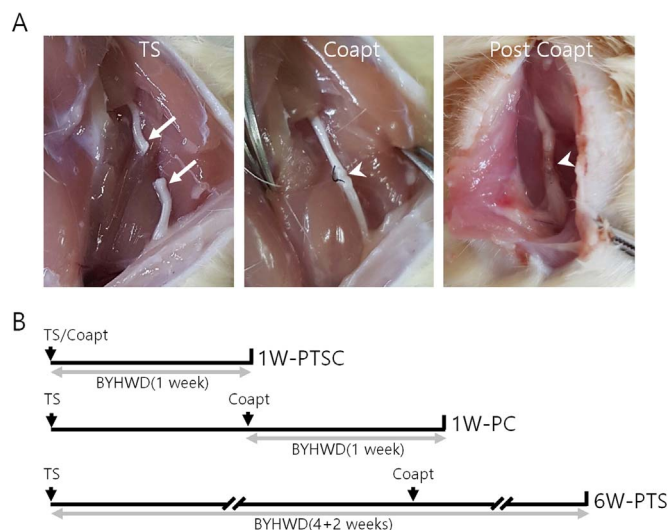


Fig. 1. Preparation of NTC animals and experimental groups. (A) Photographs showing the sciatic nerves after transection (TS; left) (proximal and distal cut ends marked by upper and lower arrows respectively) and immediately after cross-coaptation (Coapt; middle), and 2 weeks later (right). (areas of coaptation marked by arrowheads). (B) Preparation of experimental animal groups used in the present study. Animals were given nerve transection, coaptation (Coapt), and oral administration of BYHWD (see Materials and Methods for details).

outgrowth.

2.2. Experimental animals and surgery

Sprague-Dawley rats (male, 200–250 g, DaeHan Biolink, Seoul, Korea) and mice (male, 25–30 g, DaeHan Biolink, Seoul, Korea) were maintained in animal rooms with regulated temperature (22°C), 60% humidity, and a 12-h light and 12-h dark cycle. Animals were fed commercial chow (DaeHan Biolink) and allowed to drink water ad libitum. All protocols involving live and postoperative animal care were approved by the Daejeon University Institutional Animal Use and Care Committee and in accordance with the Animal-Use Statement and Ethics Committee Approval Statement for Animal Experiments provided by Daejeon University (Protocol number: DJUAR2014-036; Daejeon, Korea). For the experiments using NTC animals, rats were anesthetized with a mixed dosage of ketamine (80 mg/kg) and xylazine (5 mg/kg). For nerve transection, the sciatic nerve exposed on the middle thigh was cut, and the proximal and distal ends of the cut nerves were coapted immediately, or sutured to adjacent muscle tissues for a later coaptation surgery (Fig. 1A). The skin at the surgery area was sutured and animal was returned to the animal room after recovery from narcosis. For cross-coaptation of transected nerves, the surgery area was reopened 1 or 4 weeks after initial surgery and both transected ends were removed carefully from the sutured muscle, aligned and coapted each other by using surgery string (10-0 nylon for mice, 9-0 for rats). BYHWD (400 mg/kg body weight; 200 mg/ml saline solution) or an equivalent volume of saline was orally administered on a daily basis to NTC animals. In our previous study, we determined that a dose of 400 mg/kg was optimal to induce regenerative responses of axons after sciatic nerves injury (Chang et al., 2016), and thus used the same dose for current investigation. Based on the time schedule of surgery and drug injection, following experimental groups were established; (i) 1 week-BYHWD treatment post transection and coaptation (1W-PTSC), (ii) 1 week-BYHWD treatment post coaptation (1W-PC), (iii) 6-week BYHWD treatment post transection (6W-PTS) (Fig. 1B). All sciatic nerve surgeries in animals were given unilaterally on the left side of the hind leg.

2.3. Western blot analysis

Proximal and distal stumps (approximately 0.7 cm length) of the sciatic nerve were prepared from rats and suspended in 100–200 µl of triton lysis buffer which is composed of 20 mM Tris, pH 7.4, 137 mM NaCl, 25 mM β-glycerophosphate, pH 7.14, 2 mM sodium pyrophosphate, 2 mM ethylenediaminetetraacetic acid (EDTA), 1 mM Na₃VO₄ (Sigma-Aldrich), 1% Triton X-100, 10% glycerol, 5 µg/ml leupeptin (Sigma-Aldrich), 5 µg/ml aprotinin (Sigma-Aldrich), 3 µM benzamidin (Sigma-Aldrich), 0.5 mM dithiothreitol (DTT, Sigma-Aldrich), 1 mM phenylmethanesulfonyl fluoride (PMSF; Sigma-Aldrich) and was sonicated. The supernatant was taken after centrifugation at 8050 g for 10 min at 4 °C. Protein (15 µg) was used for SDS-polyacrylamide gel electrophoresis and immunoblotting with monoclonal anti-phospho-Erk1/2 antibody (1:2000, Cell Signaling Technology, Danvers, USA), monoclonal anti-cell division cycle 2 (Cdc2) antibody (1:2000, Santa Cruz Biotechnology Inc., Dallas, USA), monoclonal anti-phospho-vimentin antibody (1:2000, MBL, Woburn, USA) that binds specifically to vimentin (serine 55) phosphorylated by Cdc2 kinase (Tsujimura et al., 1994), monoclonal anti-actin antibody (1:10,000, MP Bio, Aurora, USA), and horse radish peroxidase (HRP)-conjugated secondary antibodies (1:3000, goat anti-mouse; Santa Cruz Biotechnology). Densitometric analysis of protein bands in the X-ray film was determined using the i-Solution software (Image & Microscope Technology, Daejeon, Korea).

2.4. Immunofluorescence staining

Sciatic nerve sections (20 µm) were fixed with 4% paraformaldehyde and 4% sucrose in phosphate buffered saline (PBS) at room temperature for 40 min, permeabilized with 0.5% nonidet P-40 in PBS, and blocked with 2.5% horse serum and 2.5% bovine serum albumin for 16 h at room temperature. Staining procedure was performed by incubating nerve sections with primary antibodies including polyclonal SCG10 antibody (1:800, Novus, St. Louis, USA) and monoclonal anti-NF-200 antibody (1:400, Sigma-Aldrich) followed by fluorescein-goat anti-mouse (1:400, Molecular probes, Eugene, USA) and rhodamine-goat anti-rabbit secondary antibodies (1:400, Invitrogen, Carlsbad, USA) in 2.5% horse serum and 2.5% bovine serum albumin for 90 min at room temperature. Nerve sections were cover-slipped with gelatin mount medium, and were viewed with a Nikon fluorescence microscope (E-600). The images were captured by Nikon camera. The merged images were produced using layer-blending mode options of the Adobe Photoshop.

2.5. Retrograde tracing of DRG neurons

For retrograde tracing of the sensory neurons in the DRG and of the motor neurons in the spinal cord, rats were anesthetized with ketamine and xylazine and a red fluorescent lipophilic carbocyanine dye 1,1'-diiododecyl-3,3',3''-tetramethylindocarbocyanine perchlorate (DiI; 1 µl of 0.5% in dimethylsulfoxide, Sigma-Aldrich) was injected into the area 10 mm distal to the transection site in the sciatic nerve by using an 1 ml insulin syringe (BD Ultra-Fine, Franklin Lakes, USA). Equal amount of DiI was also injected into the same location of the sciatic nerve in intact control animals. Animals were recovered from narcosis and sacrificed 7 days after DiI injection. DRG sections at the levels of lumbar 4 – 6 and horizontal spinal cord sections below the middle thoracic level were prepared to observe DiI-labeled neurons. DiI-labeled neurons from nonconsecutive sections (20 µm thick) were counted and averaged, and the mean values of labeled neurons were compared among experimental groups.

2.6. Primary neuron culture and coculture with Schwann cells

For DRG neuron culture, DRG at lumbar 4 – 6 levels were prepared

from rats and dissociated with 0.5 mg/ml type XI collagenase (Sigma, USA) in BME for 90 min at 37 °C. After washing twice with Basal Medium Eagle (BME), cells were treated with 5 µg/ml type SII trypsin for 15 min and followed by inhibition reaction for 5 min in 50 µg/ml of soybean trypsin inhibitor (Sigma-Aldrich), 0.5 mM EDTA, and 20 µg/ml of DNase I. Neurons (1.5×10^2 cells in 24-well plate) in BME containing 10% serum composed of 5% fetal bovine serum plus 5% horse serum and 2 mM glutamine and 1% penicillin-streptomycin were plated on 12-mm coverslips (Marlenfeld GmbH & Co., Lauda-Königshofen, Germany) precoated with poly-L-ornithine (0.1 mg/ml; Sigma-Aldrich) and laminin (0.02 mg/ml; BD Bioscience, San Diego, USA). Cells were cultured for 24 h in the presence of cytosine-β-D-arabinofuranoside (AraC, 2.5 mM; Sigma-Aldrich) to remove proliferating non-neuronal cells. For a coculture of DRG neurons and Schwann cells, Schwann cells were prepared from the distal sciatic nerves (approximately 1 cm length) in rats which had been transected for 4 weeks. The epineurium of the nerve was stripped off and the nerve was cut into small pieces. Schwann cells were dissociated similarly as DRG neurons above, and Schwann cells (1×10^4) were added to AraC-treated DRG neuron culture. Cells were incubated at 37 °C for 48 h prior to cell fixation for immunofluorescence staining. The culture was treated with BYHWD (0.5 mg/ml) or equivalent volume of saline for 48 h prior to cell fixation. After immunofluorescence staining with antibodies against neurofilament-200 (NF-200) (1:400, mouse monoclonal, Sigma) and GAP-43 (1:400, rabbit polyclonal, Santa Cruz Biotechnology), digital images were captured and transferred to the Adobe Photoshop Program. The length of neurite processes was analyzed by i-Solutions of software program (Image & Microscope Technology).

2.7. Behavioral tests

Rats and mice in 6W-PTS group together with untreated control animals were subjected to hot plate and tread mill training tests. Hot plate test was performed similarly as described previously (Ahn et al., 2016). Briefly, animals were adapted for 10 min on the surface on a plate which was adjusted to 30 °C, and transferred to the hot plate adjusted to 60 °C for 30 s. The rat on the plate was covered with a beaker and the latency to the lifting response of the hind paw was determined by analyzing real time images of animal movement. Motor activity in mice was examined by measuring the running scores on a treadmill equipment (Youngsung Biotech, Seoul, Korea). Animals were placed on a treadmill and adjusted for 10 min. The running speed was adjusted to 12 m/min and the electricity (100 V) was applied to the metal mesh set to the end of the running pad. The time that the animal was maintained on the pad without falling off was measured. In both tests above, behavioral scores were averaged by conducting the tests three times with a 10 min interval.

2.8. Statistical analysis

Data were presented as mean ± standard error of mean (SEM). The number data in individual groups were compared by one-way ANOVA followed by Tukey test post-hoc analysis (SPSS computer software version 21.0), and statistically significant differences were reported as **P* < 0.05, ***P* < 0.01, ****P* < 0.001.

3. Results

We first examined whether the axons regrew through reconnected distal nerve stump in NTC animals. One week after NTC, regenerating axons were clearly identified in the distal nerve sections by immunofluorescence staining of NF-200 proteins (Fig. 2A). SCG10, which was used as another marker protein detecting regenerating axons showed similar staining pattern as NF-200. Compared with 1W-PTSC group, axonal regrowth through distal nerve was retarded in 1W-PC group animals in which nerve coaptation was performed one week after

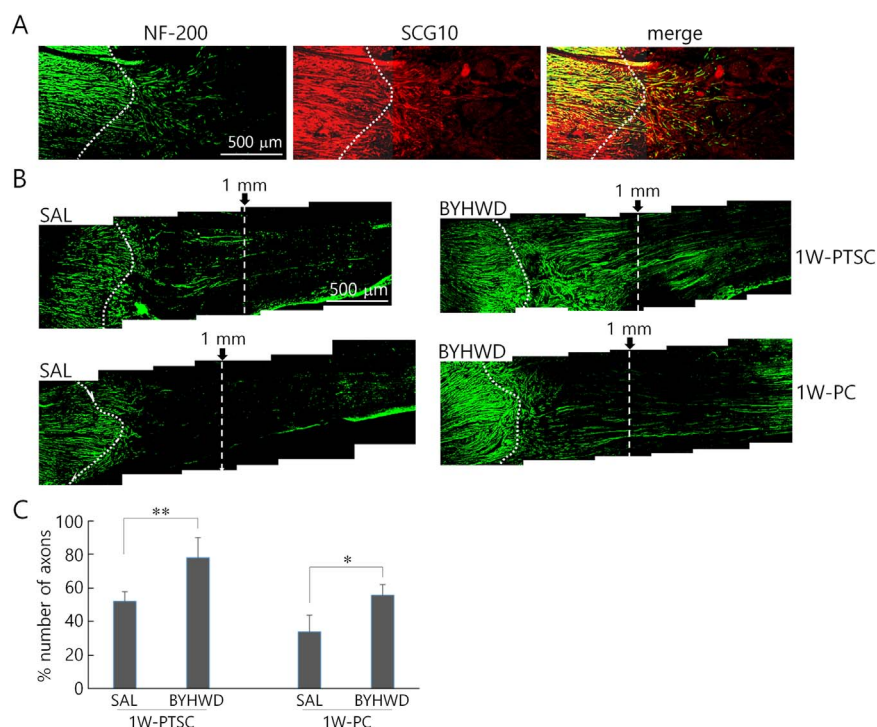


Fig. 2. Enhancement of axonal regeneration in NTC animals by BYHWD administration. (A) Identification of regenerating axons in nerve sections. One week after NTC, longitudinal nerve sections were prepared around the surgery area and used for immunofluorescence staining for NF-200 and SCG10 proteins. Dotted curves denote transection area. (B) Effects of BYHWD treatment on axonal regeneration. BYHWD or saline was treated in 1W-PTSC and 1W-PC animals, and longitudinal nerve sections were subjected to immunofluorescence staining of NF-200. In (A) and (B), multiple images were aligned into a single continuous image. (C) Quantification of axonal regeneration. The number of axons crossing the vertical line at 1 mm location (marked by vertical arrows in B) distal to the transection area (dotted lines in B) was counted in the longitudinal nerve section and a percentage of the number of axons relative to the number at the location 300 μ m proximal to the transection area was determined. At least 5 nonconsecutive sections from each animal were averaged and compared among experimental groups. N (number of animals) = 4. * $p < 0.05$, ** $p < 0.01$ (One-Way ANOVA).

transection (Figs. 1B and 2B). Then, axonal regeneration was significantly improved by BYHWD treatment in both 1W-PTSC and 1W-PC groups (Fig. 2C).

To address whether axonal regeneration enhanced by BYHWD treatment involved activation of Schwann cells or regenerating axon itself, we analyzed changes in levels of Erk1/2 activation and Cdc2 phosphorylation of vimentin in the injured sciatic nerves. In 1W-PTSC animals, phospho-Erk1/2 were induced in both proximal and distal nerve stumps, and the protein signals were largely shifted to the distal stump after BYHWD treatment (Fig. 3A). However, in 1W-PC group, phospho-Erk1/2 was induced almost exclusively in the proximal nerve in both saline and BYHWD groups, indicating that the distal induction of Erk1/2 activity was less responsive by BYHWD when the time interval between transection and coaptation was extended by 1 week. To further investigate whether BYHWD treatment affected the activation of Schwann cells in NTC animals, we analyzed the production of Schwann cell protein Cdc2 kinase and its phosphorylation of vimentin in the injured sciatic nerve. Cdc2 was clearly induced in the proximal and distal portions of the transected nerve in 1W-PTSC animals, but induction levels were much lower in 1W-PC compared to 1W-PTSC (Fig. 3B). After BYHWD treatment, Cdc2 levels were increased both in the proximal and distal nerves of 1W-PC animals while in 1W-PTSC animals, Cdc2 was distributed almost exclusively in the distal nerve. Vimentin phosphorylation was upregulated in a similar manner as Cdc2 protein in 1W-PTSC and 1W-PC groups (Fig. 3C), indicating that Cdc2 and its kinase activity were similarly regulated in NTC animals.

Since we observed that delayed coaptation restricted the effects of BYHWD on axonal regeneration, we designed an experiment in which BYHWD was administered through the whole 6 week-period of transection including post-coaptation (6W-PTS in Fig. 1B). Here, phospho-Erk1/2 levels were higher in the distal nerve compared to proximal nerve and were greatly upregulated by BYHWD treatment in both proximal and distal nerves (Fig. 4A). Axonal extension through the reconnected distal nerve was gradually decreased when measured 1 mm and 5 mm locations distal to the injury site, but the administration of BYHWD significantly improved axonal regeneration at these locations (Fig. 4B, C). To further confirm the effects of BYHWD on axonal regeneration in the sciatic sensory and motor nerves, we

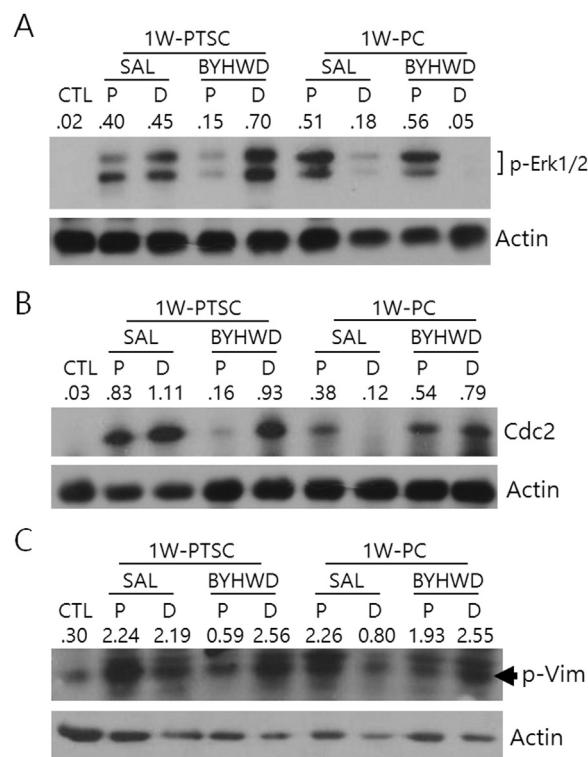


Fig. 3. Effects of BYHWD on activation of Erk1/2 and Cdc2 proteins in the proximal (P) and distal (D) portions of the sciatic nerve in NTC animals. Nerve segments prepared from 1W-PTSC and 1W-PC animals were used for western blot analysis for phospho-Erk1/2 (A), Cdc2 (B), and phospho-vimentin (C). Western blotting for actin was used as an internal loading control. Numbers above the western blot image indicate the band intensity of the proteins relative to actin control.

analyzed regenerative responses of axons by the retrograde tracing technique. Following a 4 week-period of transection, DiI was injected into the nerve 1 week after coaptation, and DiI-labeled neurons were analyzed 1 week later (Fig. 5A). Retrograde labeling of DRG sensory

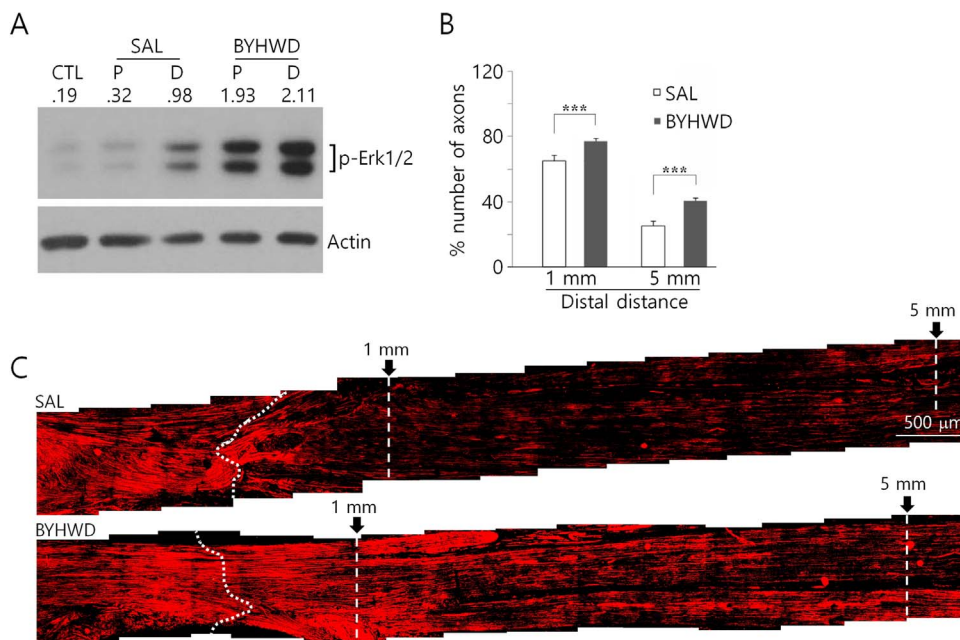


Fig. 4. Effects of BYHWD on axonal regeneration in 6W-PTS animals. The sciatic nerve in rats was cut for 4 weeks and reconnected for 2 weeks. BYHWD or saline was administered during 6-week period. (A) Western blot analysis of phospho-Erk1/2 in the proximal (P) and distal (D) portions of the nerve. Western blotting for actin was used as an internal loading control. Numbers above the western blot image indicate the band intensity of phospho-Erk1/2 relative to actin control. (B) Quantitative comparison of regenerating axons between saline- and BYHWD-treated animals. The number of axons crossing the vertical line at 1 mm and 5 mm locations (marked by filled vertical arrows in C) distal to the transection area (dotted curves in C) was counted in the longitudinal nerve section and a percentage of the number of axons relative to the number at the location 300 μ m proximal to the transection area was determined. At least, 5 nonconsecutive sections were analyzed and averaged. N (number of animals) = 4. *** p < 0.001 (One-Way ANOVA). (C) Representative immunofluorescence images showing regenerating axons in the longitudinal nerve sections. Multiple images were aligned into a single continuous image.

Fig 4 Rev

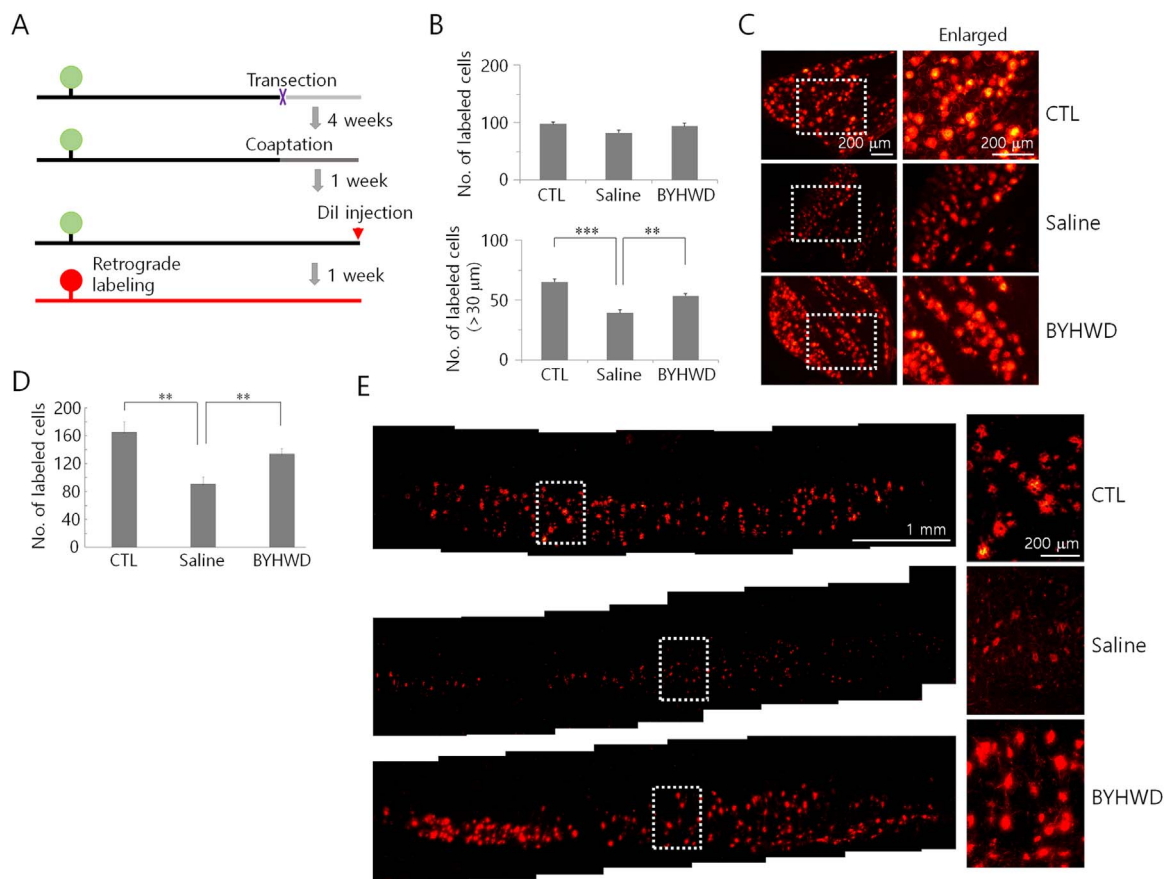


Fig. 5. Retrograde tracing and quantification of regenerating axons in 6W-PTS animals. (A) Schematic diagram of experimental procedure. Four weeks after transection, the nerve was coapted, and 1 week later, Dil was injected at the location 1 cm distal to the transection site. One week later, DRG and horizontal spinal cord sections were prepared for microscopic analysis of Dil-labeled neurons. BYHWD or saline was orally administered for a 6 week period. Dil was injected into the intact control animals (CTL) at the same location for 1 week, as well. (B) Quantitative comparison of Dil-labeled neurons in the DRG among experimental groups. In each DRG, all of labeled neurons (upper graph) or selected neurons (larger than 30 μ m; lower graph) were counted and averaged for 5 nonconsecutive images per DRG, and the mean values of labeled neurons were compared among experimental groups. (C) Representative images of labeled neurons in the DRG. Enlarged views for dotted rectangles are shown on the right. (D) Quantification of Dil-labeled neurons in the spinal cord. The number of labeled neurons was counted and averaged for 5 nonconsecutive images of horizontal spinal cord sections and the mean values of labeled neurons were compared among experimental groups. (E) Representative images showing Dil-labeled spinal motor neurons. Multiple images were aligned into a single continuous image, and enlarged views for dotted rectangles are shown on the right. In (B) and (D), ** p < 0.01, *** p < 0.001 (One-way ANOVA, number of animals = 4).

neurons revealed that BYHWD treatment increased the levels of regenerating axons, which was not however statistically significant (upper plot in Fig. 5B). Having noted that the intensity of labeled individual neurons was much stronger in BYHWD-treated animals than saline control (see representative images in Fig. 5C), we analyzed DiI-labeled neurons whose diameter was larger than 30 μm . Quantitative comparison showed that the number of labeled neurons in BYHWD-treated group was significantly higher than that in the saline control (lower plot in Fig. 5B). Our data further showed that the number of DiI-labeled spinal motor neurons, which were identified in the spinal cord at lower thoracic levels, was significantly increased by BYHWD treatment (Fig. 5D). It was also noted that the labeling intensity of individual neurons were clearly stronger in BYHWD-treated groups compared to saline control, suggesting the qualitative enhancement of axonal regeneration in individual neurons by BYHWD (Fig. 5E).

To investigate whether Schwann cell activity was involved in the process of BYHWD-mediated axonal regeneration, we prepared and analyzed neurite outgrowth in DRG neurons which were cocultured with Schwann cells prepared from transected sciatic nerves. Neurite outgrowth was similar between cocultured DRG neurons and single cultured neurons. Then, the addition of BYHWD significantly increased neurite outgrowth of cocultured neurons (Fig. 6A, B). Signals of axonal growth-associated protein GAP-43 were detected in neuronal cell body area, and, in neurons treated with BYHWD, additional signals were detected at the location of neuritic tips (Fig. 6C).

Since BYHWD clearly facilitated axonal regeneration in 6W-PTS animals, we asked if BYHWD administration also improved behavioral recovery. In a thermal sensitivity test using a hot plate, a time interval displaying the initial withdrawal response lifting hind legs was greatly delayed in 6W-PTS animals compared to control animals (Fig. 7A). Then, the administration of BYHWD significantly improved the time response. We also analyzed motor activity by measuring the time period that the animals ran on the treadmill without falling off. Mean running time in 6W-PTS group, which was markedly decreased compared to control animals, was significantly improved by BYHWD treatment (Fig. 7B).

As an initial approach to examining the effects of individual herbal constituents of BYHWD on regenerative responses, we investigated neurite outgrowth of DRG neurons which were prepared from injured sciatic nerve. Mean neurite length showed no significant difference among drug- and saline-treated groups (data not shown). However, a comparison of mean values of neurite length which was longer than 500 μm revealed that the neurite outgrowth in cell groups treated with

Radix Paeoniae Rubra (RPR) and Radix Angelicae Sinensis (RAS) were significantly enhanced compared to saline-treated group (Fig. 8A). Similarly, mean length of neurites in cell groups treated with amygdalin, calycosin, and kaempferol, chemical components which were extracted from herbal constituents of BYHWD, was not significantly different from saline-treated control (data not shown). Here, too, a comparison of neurite length longer than 500 μm between drug-treated and saline control groups revealed that mean neurite length was significantly increased in all three groups treated with amygdalin, calycosin, and kaempferol than saline control (Fig. 8B).

4. Discussion

Although the axons in the peripheral nerves can regenerate after injury, a degree of functional recovery is variable depending on the injury types, pathophysiological status of injury area, and species difference. Recently we have reported that BYHWD administration improved axonal regeneration in the sciatic nerve after crush injury (Chang et al., 2016), and here we extended the investigation on its efficacy into the system of nerve transection. Daily administration of BYHWD promoted axonal regeneration and improved functional recovery in the reconnected nerves after prolonged denervation, raising the possibility that BYHWD may be considered as a therapeutic target for nerve repair.

Our data showed that the distal elongation of transected axons was attenuated by delaying before resuture, compared to immediate coaptation. After peripheral nerve injury, Schwann cells are released from degenerating axons and subsequently dedifferentiate, proliferate and redifferentiate to remyelinate regenerating axons (Chen et al., 2007; Namgung, 2014; Jessen et al., 2015). Schwann cell-specific induction of Erk1/2 signaling pathway in mice resulted in Schwann cell dedifferentiation and subsequent inflammatory response (Napoli et al., 2012). Thus, it can be inferred that immediate reconnection of cut nerve may facilitate the plastic responsiveness of distal Schwann cells via the interaction with regrowing axons (Arthur-Farraj et al., 2012), whereas prolonged disconnection could alter the phenotypic responsiveness of Schwann cells rendering them less supportive for axonal regeneration. Administration of BYHWD after nerve coaptation significantly improved the distal elongation of axons in both 1W-PTSC and 1W-PC groups, suggesting that BYHWD may not only accelerate the distal elongation of regenerating axons as an intrinsic factor, but also activate distal Schwann cells.

Previously, we found that Cdc2 kinase is strongly but transiently

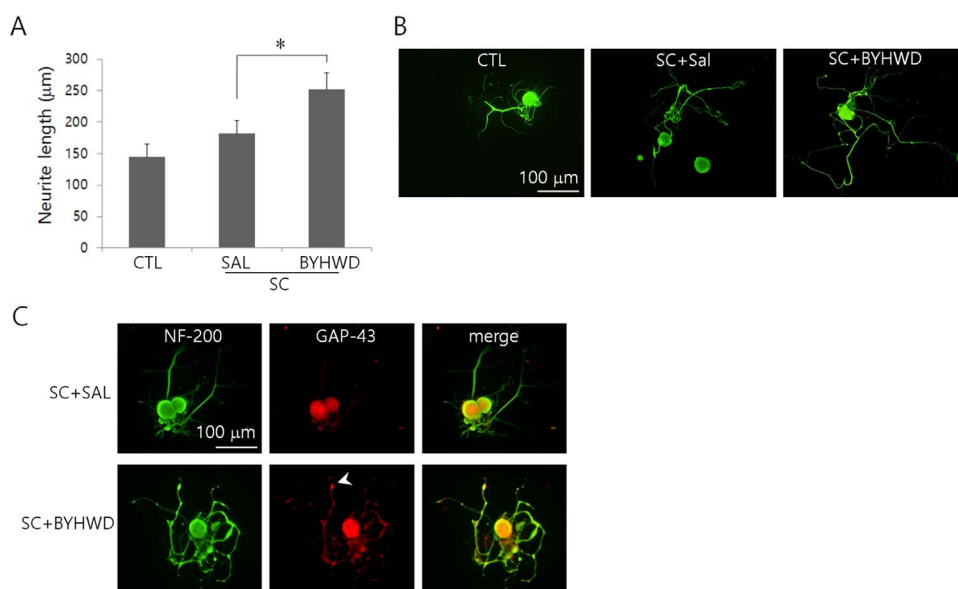


Fig. 6. Effects of BYHWD on the neurite outgrowth in DRG neurons. Schwann cells were prepared from the distal sciatic nerves of animals which had been transected for 4 weeks. DRG neurons were treated with AraC and further cultured alone or cocultured with Schwann cells after adding BYHWD (0.3 mg/ml) or saline for 48 h. (A) Quantification of neurite length of DRG neurons in single culture (CTL) or coculture with Schwann cells (SC). Ten to fifteen neurons were randomly selected from each culture, and neurite length was measured, averaged, and compared among experimental groups. * $p < 0.05$, N (number of independent experiments = 4, One-way ANOVA). (B) Representative immunofluorescence images of NF-200-stained neurons. (C) Induction of GAP-43 signals in DRG neurons. GAP-43 signals were observed in the neuritic tips as well as cell body area of BYHWD-treated DRG neurons which were cocultured with Schwann cells (marked arrowhead).

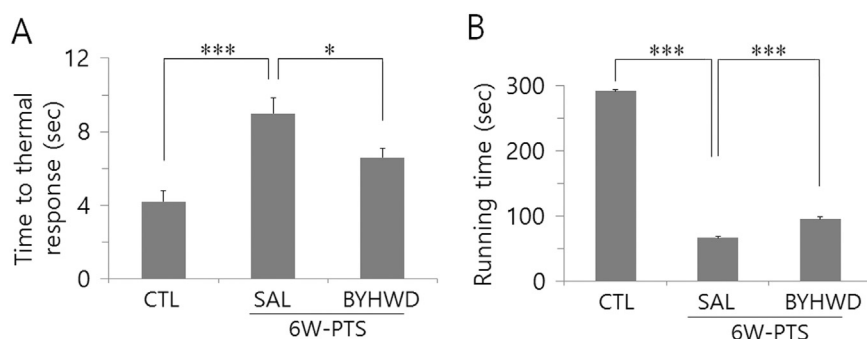


Fig. 7. Improvement of behavioral scores in 6W-PTS animals after BYHWD treatment. 6W-PTS animals were treated with either BYHWD or saline for 6 weeks, and, together with intact rats (CTL), were subjected to behavioral tests. (A) Hot plate test to measure the thermal sensitivity. (B) Treadmill running test. *p < 0.05, ***p < 0.001. (One-way ANOVA, number of animals per group = 5).

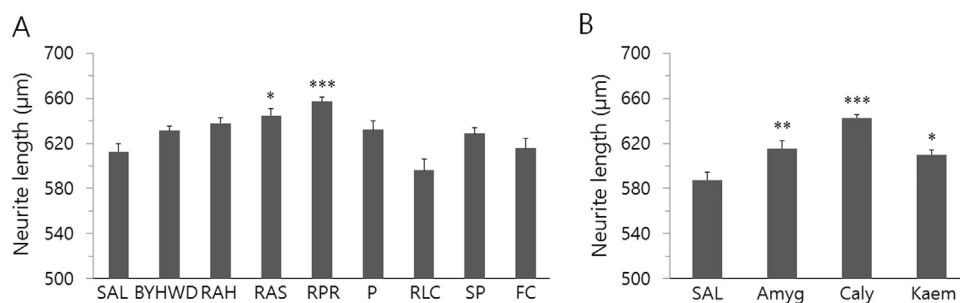


Fig. 8. Comparison of neurite outgrowth in DRG neurons treated with herbal constituents and chemical ingredients of BYHWD. Three days after sciatic nerve injury, DRG neurons were prepared from rats and treated with individual herbal drugs (0.3 mg/ml; A) and chemical compounds (15 μM) for 48 h. Neuritic processes were visualized by immunofluorescence staining with NF-200. In each neuron, neurite length longer than 500 μm was selected and mean length was determined. Fifteen neurons in (A) and twenty neurons in (B) were averaged per each group and compared. *P < 0.05, **P < 0.01, ***P < 0.001 vs. Saline control (One-way ANOVA).

In (A), SAL; saline control, RAH; Radix Astragali seu Hedysari, RAS; Radix Angelicae Sinensis, RPR; Radix Paeoniae Rubra, P; Pheretima, RLC; Rhizoma Ligustici Chuanxiong, SP; Semen Persicae, FC; Flos Carthami, and in (B), Amyg; amygdalin, Caly; calycosin, Kaem; kaempferol.

induced from Schwann cells in the injured peripheral nerve and mediates the distal migration of Schwann cells (Han et al., 2007). Furthermore, Cdc2 phosphorylates vimentin and mediates the activation of β1 integrin as a process of Schwann cell interaction with regenerating axons (Chang et al., 2012; Namgung, 2014). Here, our data showed that Cdc2 and phospho-vimentin were almost exclusively present in the distal stump after BYHWD treatment in 1W-PTSC group, and the same proteins were newly induced by BYHWD in the distal nerve of 1W-PC animals. BYHWD treatment in 1W-PTSC group may facilitate distal migration of Schwann cells bearing newly expressing Cdc2, as we have demonstrated similarly in the sciatic nerve after crush injury (Han et al., 2007). Alternatively, BYHWD could downregulate Cdc2 activity in the proximal Schwann cells. However, in 1W-PC group, Schwann cells bearing Cdc2 activity, which had been induced and accumulated in the proximal Schwann cells, might be fully activated and less plastic than those in 1W-PTSC animals, and thus would not be affected by BYHWD; instead, BYHWD could increase Cdc2 activity in the distal Schwann cells after reconnection. Taken together, our data suggest that BYHWD may be involved in activating Schwann cells and support axonal regeneration in the transected nerves.

Next, we asked if BYHWD was supportive for axonal regeneration in chronically denervated nerves. First, it was noted that in 6W-PTSC animals, phospho-Erk1/2 signals were much stronger in BYHWD-treated group than saline control. Phospho-Erk1/2 is known to be induced from regenerating axons and mediates retrograde transport of the lesion signals to the neuronal cell body (Perlson et al., 2005; Johanson et al., 1995). Phospho-Erk1/2 is also expressed from Schwann cells in the injured nerves and functionally involved in axonal regeneration (Agthong et al., 2006; Seo et al., 2009). Thus, Erk1/2 activity, which is upregulated by BYHWD treatment in both Schwann cells and regenerating axons, may contribute to enhanced axonal regeneration. A quantitative comparison of the number of regenerating axons of DRG sensory neurons revealed that the total number of retrogradely labeled neurons after nerve injury were close to that of untreated control and no significant difference was seen between saline and BYHWD-treated animals. When we selectively analyzed larger neurons, BYHWD treatment was effective in increasing the regeneration of corresponding axons. In the DRG, larger mechanosensory neurons with

myelinated thick Aα and Aβ fibers are responsible for touch sensation while smaller neurons with thin Aδ fiber and unmyelinated c-fiber transmit pain (Gardner and Johnson, 2013). Our data show that the rate of axonal regeneration was retarded in larger neurons compared to smaller ones, and BYHWD facilitated their regeneration. Perhaps, large myelinated axons such as Aα and Aβ fiber may need more time to regenerate than small fibers, and BYHWD facilitated regeneration processes of larger fibers. Our data further show that the levels of axonal regeneration in terms of relative numbers of DiI-labeled neurons in 6W-PTSC animals to intact control was lower in spinal motor neurons than DRG neurons, which is consistent with previous data on distinct regenerative responses in sensory and motor axons (Suzuki et al., 1998; Negredo et al., 2004). Since the cell bodies of motor neurons are located within the spinal cord, an environment of central nervous system, intrinsic regeneration capability of motor axons may be lower than DRG neurons (Chong et al., 1999; Fawcett et al., 2001). Also, a physical distance for retrograde tracer to reach the cell body of spinal motor neurons is longer than DRG neurons. BYHWD treatment not only elevated the number of retrograde labeled-neurons but also enhanced the labeling intensity of individual neurons compared to saline control. Thus, BYHWD may play a role in facilitating axonal regeneration in both sciatic sensory and motor axons after chronic denervation.

Previously we have reported that Schwann cells in the sciatic nerve given crush injury are activated and support axonal regeneration (Chang et al., 2012). However, after chronic denervation, distal Schwann cells may deteriorate the responsiveness to regenerating axons. Indeed, primary DRG neurons cocultured with Schwann cells from chronically denervated nerves did not show any significant increases in neurite outgrowth compared to DRG neuron single culture. Then, BYHWD treatment enhanced neurite outgrowth of DRG neurons, and these neurons showed the induction of GAP-43 protein signals at the neuritic processes. GAP-43 is an axonal growth-associate protein which is induced at gene expression level after nerve injury and thus has been used as an intrinsic regeneration marker of axonal regeneration (Skene, 1989; Smith and Skene, 1997). Since DRG neurons were prepared from intact animals and cultured after eliminating proliferating nonneuronal cells, GAP-43 production and its effect on neurite outgrowth could be minimal in this condition. Here, the

addition of Schwann cells from chronically transected sciatic nerve was not effective in inducing GAP-43 and neurite outgrowth, but BYHWD treatment improved the neuronal responses. Thus, these results strongly suggest that, in culture system, BYHWD may act directly on neurons and potentiate its regenerative responses. These interpretation is consistent with our previous report showing the effects of BYHWD on DRG neurons from preconditioned nerves (Chang et al., 2016).

To determine which herbal components of BYHWD was effective in mediating regenerative responses, we analyzed neurite outgrowth of cultured DRG neurons treated with individual herbal drugs, as previously reported to screen the active herbal components from decoctions (Kim et al., 2017; Tohda et al., 2005). Here in our analysis, some herbal components such as Radix Paeoniae Rubra and Radix Angelicae Sinensis were more active than the others including BYHWD in promoting the elongation of large sized-neurites whereas a certain component (Pheretima) was less effective than BYHWD. It was previously reported that Radix Paeoniae Rubra extract showed antitumor and antithrombotic activity (Xie et al., 2017; Lin et al., 2016), but no studies on the efficacy of herbal constituents of BYHWD on axonal regeneration have been reported so far, and our data represents a new evidence.

We also investigated some chemical ingredients of BYHWD such as amygdalin, calycosin, and kaempferol because they have been investigated in relation to neuroprotective activities in cultured neurons (Yu et al., 2005; E.J. Yang et al., 2014; C. Yang et al., 2014). Treatment of these drugs into preinjured DRG neurons facilitated neurite outgrowth, and it was particularly noted that calycosin, which was shown to protect brain from ischemic injury and cognitive impairment (Guo et al., 2017; Wang and Zhao, 2016), induced neurite outgrowth most efficiently. Calycosin is extracted from Radix Astragali seu Hedysari, a most abundant herbal component of BYHWD accounting 84% in dry weight, raising a possibility that it may play a predominant role in regenerative responses in the injured peripheral nerves. Further studies on its activity on axonal regeneration after nerve transection may be of great importance to elucidate mechanistic basis on BYHWD activity.

5. Conclusion

Our data suggest that in chronic denervation, BYHWD may enhance regenerative responses both in regrowing axons and Schwann cells. Specifically, BYHWD treatment upregulated the production of GAP-43 in axons and also activated Cdc2 signaling pathway in Schwann cells. In rats treated BYHWD for a prolonged period (i.e., 6W-PTS group), behavioral scores for sensory and motor activity were significantly improved, implicating the potential therapeutic applicability of BYHWD to human patients suffering from peripheral nerve injury. In human, an efficacy of axonal regeneration after peripheral nerve injury is much lower than experimental animals such as rats and mice. This could be partly due to the fact that physical distance of regenerating axons to the target tissues is much longer and accordingly, Schwann cells in the denervated distal nerve may become less responsive to axons after prolonged denervation. In this situation, BYHWD treatment may accelerate axonal regeneration rate, activate atrophied distal Schwann cells, and thereby contribute to timely interaction between Schwann cells and regenerating axons and functional recovery. More intensive studies on compositional analysis of BYHWD will be critical to gain insights into mechanistic basis on its action and for the development of therapeutic strategies.

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